

ENERGY PERCEPTION VS. REALITY IN FRANCE AND GERMANY

Compare public perception with actual energy production and emissions data in France and Germany, using Reddit discussions and official datasets. The study aims to identify discrepancies, analyze public opinion across energy types, and provide insights through data visualization.

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PROJECT PRESENTATION

- Overview:**
- Comparison of public perception vs. actual data on energy production and emissions in France and Germany.
 - Analysis of Reddit discussions and official energy datasets to identify discrepancies.

- Key Highlights:**
- Sentiment analysis of Reddit discussions on energy topics.
 - Integration of official energy production and emissions data.
 - Data visualization to provide actionable insights.

- Research Questions:**
- Key differences between public perception and actual energy production data in France and Germany.
 - Sentiment variation across different energy types (e.g., nuclear, renewables).
 - Alignment between public opinion and emissions realities.

STACK



ANALYSIS AND LIMITATIONS

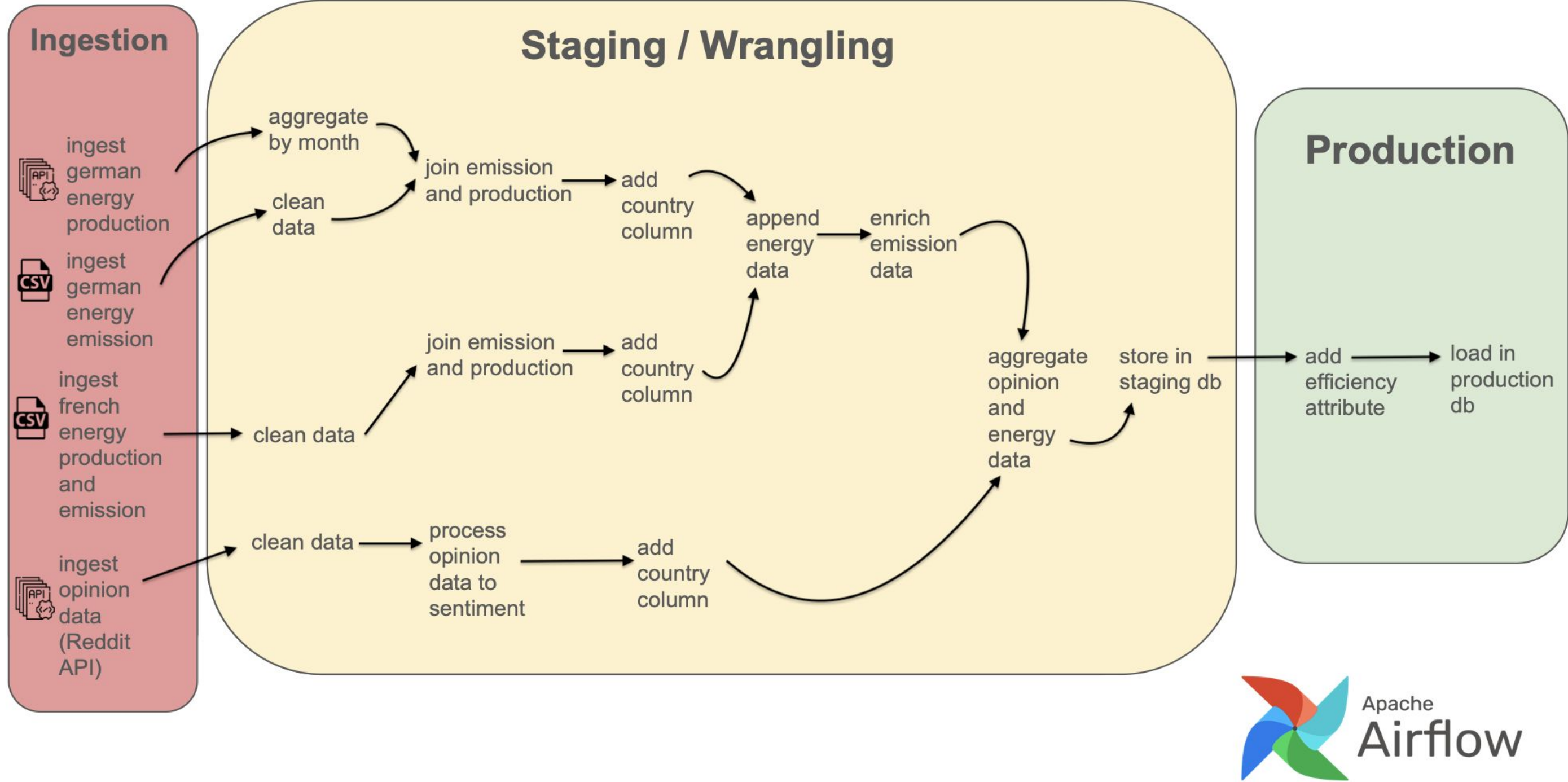
- Key Insights**
- Nuclear Energy:**
 - Negative sentiment in Germany aligns with nuclear phase-out policy.
 - Positive sentiment in France reflects reliance on nuclear energy.
 - Renewables:** Positively perceived in both countries, consistent with global green energy trends.
- Limitations**
- Temporal Gap: Limited public opinion data before 2018.
 - Sampling Bias: Reddit comments, often negative, may skew sentiment analysis.
 - Results may not represent the broader population due to Reddit-specific biases.

DATA SOURCE

- Public Opinion Data**
- Source:** Reddit API
 - Description:** Posts and comments extracted from energy-related subreddits in France and Germany.
 - Processing:** Sentiment classification (positive, neutral, negative) and association with specific energy topics.
- Energy Production and Emission Data**
- Germany**
 - Source:** RTE (manually downloaded CSV)
 - Data:** Energy production and emissions
 - France**
 - Sources:** [Energy-Chart API](#) (for production data).
 - Energy-Chart** (manually downloaded CSV for emission data).

PIPELINE

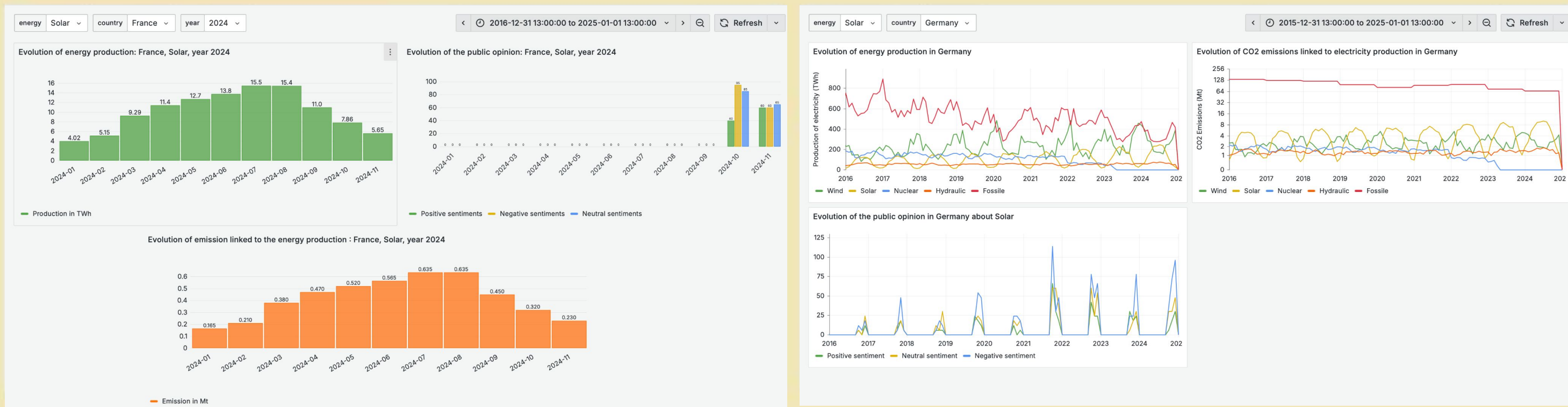
- Ingestion Phase (Landing Zone):**
- Collected Reddit posts and comments on energy topics using the Reddit API.
 - Downloaded energy production and emissions data for France (RTE) and Germany (Energy-Chart API, CSV files).
 - Stored raw data temporarily in a landing zone for future processing.
- Transformation Phase (Staging Zone):**
- Cleaned data by removing duplicates, standardizing date formats, and handling missing values.
 - Performed sentiment analysis by applying a multilingual model on Reddit data, classifying sentiment as positive, neutral, or negative.
 - Merged energy datasets with sentiment data, enriching the data for analysis.
- Production Phase (Final Database):**
- Integrated cleaned, transformed data into a star-schema database for efficient querying.
 - Stored enriched datasets with public sentiment linked to energy production and emissions.
 - Enabled long-term access to the final, processed data for reporting and visualization.



IMPROVMENT

- Get data on accidents:** to see how it can affect the public opinion and the use of the energies concerned.
- Enhance Sentiment Analysis:** Improve the model for better nuance, especially on controversial topics like nuclear energy.
- Automate Data Ingestion:** Streamline processes for manual datasets to improve scalability and efficiency.
- Expand Visualizations:** Build detailed Grafana dashboards, including regional trends for localized insights.
- Address Data Gaps:** Use advanced LLMs (e.g., Gemini) to generate synthetic opinions, fine-tuned with historical surveys, bridging pre-2018 data gaps.

VIZUALISATION



Dashboards in Grafana